THESIS INFORMATION

Dissertation title: **ABNORMAL EVENT DETECTION IN VIDEOS**

Major code: **Computer Science**

Code: **9480101**

PhD student: **TRAN MINH TUNG**

Scientific advisors: **PhD. NGUYEN TAN TRAN MINH KHANG**
and Assoc. Prof. PhD. NGUYEN VAN TAM

University: **UNIVERSITY OF INFORMATION TECHNOLOGY, VNU-HCM**

ABSTRACT

Recently, abnormal event detection in videos has become one of the main, challenging, and interesting topics for researchers in the field of computer vision and machine learning, especially in detecting anomalous events in aerial videos. This problem is widely applied in various fields, including traffic surveillance, border security, medical diagnosis, smart agriculture, delivery, disaster rescue, and environmental monitoring. This PhD dissertation focuses on studying, surveying and systematizing research methodologies and approaches in video anomaly detection, publicly available benchmark datasets, proposing a novel method for detecting abnormal events at frame level in aerial videos with traffic scenes, and constructing an new anomalous video dataset with traffic scenarios recorded by drones. In order to fulfill these objectives, the dissertation has made three main contributions, which are presented as follows:

(1) Proposing a novel method for frame-level anomaly detection in aerial traffic videos: In the first contribution, the PhD dissertation proposed a method based on Vision Transformer architecture and trained the model using unsupervised learning strategy through exploiting and integrating spatio-temporal features to predict future frames for the task of detecting anomalies in aerial videos. Furthermore, experiments were conducted and demonstrated the superior performance of the proposed method compared to current state-

of-the-art methods on two large-scale aerial video anomaly datasets, namely UIT-ADrone and Drone-Anomaly.

(2) Contribution to a comprehensive survey study on the problem of detecting anomalies in videos: As the second contribution, the PhD dissertation focused on surveying, and systematizing research approaches and techniques, public benchmark datasets, performance of current state-of-the-art methods, and popular metrics for the problem of detecting abnormal events in videos. In addition, the PhD dissertation classified anomaly problems in videos into four groups: anomaly detection; anomaly classification; anomaly prediction; and anomaly localization. Additionally, extensive experiments were conducted to compare the performance of state-of-the-art methods for detecting anomalies in videos. Moreover, the advantages and limitations of state-of-the-art methods as well as potential applications of this problem in practice are discussed in depth.

(3) Building and publishing the UIT-ADrone dataset of traffic scenes in Ho Chi Minh City, Vietnam, for the task of detecting anomalous events in aerial surveillance videos: In the third contribution, the PhD student collected and built the largest drone-recorded video dataset to date, including data collection, data statistics, and data annotation, specifically designed to detect anomalous events from an aerial perspective.

During the implementation and completion of the dissertation, the PhD student published seven scientific papers, including three articles in reputable journals (SCIE Q1-ranked journals) and four papers presented at prestigious international and national conferences in the field of computer science and information technology.

Scientific Advisors	**PhD Student**

NGUYEN TAN TRAN MINH KHANG **NGUYEN VAN TAM** **TRAN MINH TUNG**